an absolutely straight shape. Similarly, when the mountain sinks inside the hole, it can no longer keep the horizon deformed, so the horizon vibrates, sending off gravitational waves; the waves carry away the energy of the horizon's deformation, and the horizon settles down into an absolutely spherical shape.

How does this mountain-endowed implosion relate to Price's theorem? According to the laws of physics, the horizon's mountain-like protrusion *can* be converted into gravitational radiation (ripples of curvature). Price's theorem tells us, then, that the protrusion *must* be converted into gravitational waves, and that this radiation must carry the protrusion completely away. *This is the mechanism that makes the hole hairless.*

Price's theorem tells us not only how a deformed hole loses its deformation, but also how a magnetized hole loses its magnetic field (Figure 7.5). (The mechanism, in this case, was already clear before Price's theorem from a computer simulation by Werner Israel and two of his Canadian students, Vicente de la Cruz and Ted Chase.) The magnetized hole is created by the implosion of a magnetized star. Before the horizon engulfs the imploding star (Figure 7.5a), the magnetic field is firmly anchored in the star's interior; electric currents inside the star prevent the field from escaping. After the star is swallowed by the horizon (Figure 7.5b), the field can no longer feel the star's electric currents; they no longer anchor it. The field now threads the horizon, rather than the star, but the horizon is a worthless anchor. The laws of physics permit the field to turn itself into electromagnetic radiation (ripples of magnetic and electric force), and Price's theorem therefore demands that it do so (Figure 7.5c). The electromagnetic radiation flies away, partly down the hole and partly away from it, leaving the hole unmagnetized (Figure 7.5d).